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USE MATHEMATICS TO INVESTIGATE AND MONITOR TYE FINANCIAL ASPECTS OF PERSONAL, BUSINESS, NATIONAL AMD INTERNAL ISSUES

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Formative assessment

Question 1

a. Elements for effective management of financial instruments

1. Risk assessment: Understand the level of risk, return, and liquidity of each instrument before investing.

2. Diversification: Spread investments across different instruments to reduce risk.

3. Time horizon:Match the investment period to your financial goals.

4. Market conditions:Monitor interest rates, inflation, and economic factors that affect instruments.

5. Liquidity needs: Ensure some instruments can be easily converted to cash when needed.

6. Costs and fees: Consider transaction costs, management fees, and taxes that reduce returns.

7. Regulatory compliance: Follow laws and regulations governing the instruments.

8. Performance monitoring:Regularly review and rebalance the portfolio.

b. Ways to control instruments to maximize profit / minimize loss

- Stock :Diversify across sectors, use stop-loss orders to limit losses, and research company fundamentals before buying/selling.

- Bond :Ladder maturities to manage interest rate risk, buy high credit-rated bonds to reduce default risk, and hold to maturity to avoid capital loss.

- Unit trusts: Choose funds with low fees and good track record, diversify across fund types, and review fund performance regularly and switch if underperforming.

Question 2

Retail value = R200 000

Excess = R4 500

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Cost of accident = R4 000

Insurance payout = Cost of accident - Excess

= R4 000 - R4 500 = -R500

Since the cost is less than the excess, Palesa cannot claim anything. She recovers R0*. She pays the R4 000 herself.

Question 3

a. Compound Interest

Formula: A = P(1 + i)^n

P = R620 800, i = 8% = 0.08, n = 4

A = 620800(1 + 0.08)^4

A = 620800(1.08)^4

A= 620800(1.360)= R844 288

Interest = A - P = 844 288 - 620 800 = R223 488

b. Simple vs Compound interest - worked example

Say R10 000 at 10% for 2 years:

Simple interest: Interest calculated only on principal each year.

Year 1: 10 000 x 10% = R1 000

Year 2: 10 000 x 10% = R1 000

Total = R2 000. Amount = R12 000

Compound interest:Interest calculated on principal + previous interest.

Year 1: 10 000 x 10% = R1 000 → New principal R11 000

Year 2: 11 000 x 10% = R1 100

Total = R2 100. Amount = R12 100

Difference: Compound gives "interest on interest" so it grows faster.

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Question 4

Option 1: Overseas investment

1. Convert R100 000 to USD at R7.01: $100000 / 7.01 = $14 265.33

2. Earn 6% for 1 year: $14265.33 x 1.06 = $15 121.25

3. Convert back at R7.29: $15121.25 x 7.29 = R110 233.91

Option 2: Local savings at 3% *

R100000 x 1.03 = R103 000

Overseas gives R110 233.91 vs Local R103 000.

The overseas investment is better by R7 233.91 because the Rand depreciated + higher interest rate.

Question 5

a. Definitions

- Marginal cost: The additional cost of producing one more unit. MC = \Delta TC / \Delta Q

- Marginal revenue: The additional revenue from selling one more unit. MR = \Delta TR / \Delta Q

Question 7

5000 pula to Rands. Rate: R1 = 0.88 pula

So: Rands = Pula / 0.88

Rands = 5000 / 0.88 = R5 681.82

Debra has R5 681.82